Cholesterol Metabolism Gene Signatures Distinguish Estrogen-Receptor Status and Define Immune-Relevant Molecular Subtypes of Breast Cancer

An Integrated Multi-Cohort Analysis of TCGA-BRCA and METABRIC

Abstract

Purpose. Cholesterol metabolism is closely linked to estrogen-receptor (ER) biology, but its value for breast cancer classification, molecular subtyping and causal inference remains incompletely quantified.

Materials and Methods. We applied co-expression network screening, machine-learning feature selection, consensus clustering, single-cell localization, immune deconvolution (CIBERSORT/ESTIMATE), SMR/HEIDI Mendelian randomization and Bayesian colocalisation to TCGA-BRCA (n = 952) and METABRIC (n = 1,979), with classifier validation in GSE21653, GSE7390 and GSE20711 and nearest-centroid subtype mapping across all external cohorts.

Results. The strict WGCNA-by-pathway intersection was empty. A pathway-first adaptation using 119 measurable cholesterol genes yielded 11 hub genes; the ER classifier reached AUCs of 0.946 (5-fold cross-validation), 0.922 (METABRIC), 0.847 (GSE21653) and 0.903 (GSE7390). Consensus clustering defined four subtypes with divergent ER composition, immune infiltration and PAM50 correspondence. Single-cell analyses localized cholesterol biosynthesis to tumour epithelial and myeloid/endothelial compartments. Pathway-wide SMR identified FDPS as a putatively causal protective gene (P = 9.8e-7; HEIDI P = 0.026; coloc.abf PP.H4 = 0.9985), with the protective direction replicated in ER+ OncoArray analyses (P = 1.54e-5) and an inverse methylation-expression correlation (ER-adjusted partial rho = -0.187, P = 2.2e-6).

Conclusion. Cholesterol metabolism genes do not define a progression-associated co-expression module but carry a strong, cross-platform ER signature, define immune-relevant molecular subtypes and include FDPS as a putatively causal protective gene for breast cancer risk.

Keywords: Breast Neoplasms; Cholesterol; Machine Learning; Receptors, Estrogen; Tumor Microenvironment

1. Introduction

Cholesterol and its derivatives support breast cancer cell proliferation, membrane biogenesis, steroid hormone synthesis and immune regulation [1,2]. ER-positive (ER+) tumours coordinately up-regulate cholesterol biosynthetic enzymes, and cholesterol-lowering agents have been proposed as adjunct therapy [3-6]. Most studies, however, examine individual genes and rarely connect pathway-level expression to patient phenotypes in a structured, multi-layer fashion [7,8]. Recent lipid-metabolism transcriptomic studies in breast cancer have emphasised prognostic signatures and immune correlates [7,8]; the present study differs by reporting an honest null network-pathway intersection, an ER-focused classifier validated across four independent expression cohorts, and a germline causal-inference layer for FDPS.

A widely used bioinformatics template proceeds from whole-transcriptome screening (WGCNA plus a pathway gene set) to machine-learning target reduction, clinical model building, molecular subtyping, single-cell localization and Mendelian-randomization causal testing. We applied this template to breast cancer using cholesterol metabolism as the biological anchor, with TCGA-BRCA and METABRIC as discovery and external cohorts, and we report the completed evidence chain from screening to causal inference.

2. Methods

2.1 Data sources and verification The overall study design and the flow of analyses are summarised in Supplementary Figure S2. For validation analyses, PAM50 calls for TCGA-BRCA were obtained from UCSC Xena (GDC hub) [9]; FDPS HM450 methylation beta values were obtained from the cBioPortal data repository (brca_tcga_methylation_hm450); the independent expression cohorts GSE21653 (Affymetrix HG-U133 Plus 2.0, NCBI GEO with GPL570 annotation) [10] and GSE7390 (Affymetrix HG-U133A, ArrayExpress E-GEOD-7390 processed matrix and SDRF clinical data, probes mapped with hgu133a.db) [11] were used; a fourth independent Affymetrix cohort (GSE20711, NCBI GEO) was used for classifier transfer; the paired normal-tumour cohort GSE15852 (Affymetrix HG-U133A; 43 pairs) was used for normal-versus-tumour expression comparisons; the immunotherapy cohort GSE91061 (RNA-seq, melanoma anti-PD-1; 51 pre-treatment samples) was used for exploratory immunotherapy-response associations; and the second single-cell atlas GSE161529 (Mendeley Data mirror of the Pal et al. atlas) [12] was used for single-cell validation. The Affymetrix annotation was taken from the GEO GPL570 platform file. TCGA-BRCA RNA-sequencing STAR counts, methylation, mutation and clinical files were obtained from the NCI GDC portal [13] and verified against the official manifest by MD5 checksums (4,393/4,393 files); progression-free interval (PFI) was taken from TCGA-CDR [14]. METABRIC expression and clinical annotation with recurrence-free survival (RFS) were obtained from cBioPortal [15]. TCGA ER/PR/HER2 status was parsed from clinical XML; METABRIC receptor status from clinical sample files.

2.2 Cohorts and endpoints

The discovery cohort comprised 952 TCGA-BRCA patients [16] with valid PFI (122 events, 12.8%; median follow-up 2.1 years; ER status for 909). The external cohort comprised 1,979 METABRIC samples [17] with RFS (803 events, 40.6%; median follow-up 8.5 years; ER status for all).

2.3 Cholesterol metabolism gene set

A cholesterol metabolism gene set was assembled from KEGG hsa00100 (steroid biosynthesis) [18] and Gene Ontology biological-process terms GO:0008203 (cholesterol metabolic process) and GO:0006695 (cholesterol biosynthetic process) [19] using org.Hs.eg.db [20]; the union contained 120 unique genes (19 from KEGG, 107 from GO:0008203, 39 from GO:0006695).

2.4 WGCNA Expression was transformed as log2(CPM+1). The 8,000 most variable genes (median absolute deviation) passing expression filters entered weighted correlation network analysis [21,22]: soft-threshold power selection by scale-free fit, adjacency |cor|^beta, topological overlap matrix, average-linkage hierarchical clustering, module cutting with a height scan (10-40 modules of >= 20 genes) and merging of modules with eigengene correlation >= 0.80. Module eigengenes were correlated with PFI event, PFI time, ER status and stage (Pearson/Spearman; Benjamini-Hochberg FDR [23]).

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2.5 Candidate selection and machine learning Model calibration was evaluated with reliability curves (decile observed-versus-predicted ER+ fractions), the Brier score, and logistic recalibration intercept and slope; clinical utility was evaluated with decision-curve analysis (net benefit across threshold probabilities) in TCGA out-of-fold cross-validated predictions and in the external METABRIC cohort. The strict intersection of trait-associated modules with the cholesterol set was empty; the candidate pool was therefore the 119 cholesterol genes measurable in both cohorts (pathway-first adaptation, documented as an analysis deviation). Expression was z-scored with TCGA statistics. LASSO logistic regression [24] (L1, C selected by 5-fold CV over a 25-point grid) and SVM-RFE (linear SVM) [25] were applied to ER status; the intersection of their selected gene lists defined 11 hub genes. Selection stability was assessed with 300 bootstrap resamples of the LASSO fit. A signature score was computed as the sum of LASSO coefficients times z-scored expression. Performance was assessed by 5-fold CV AUC in TCGA and external AUC in METABRIC (2,000-bootstrap 95% confidence intervals); as a secondary endpoint, the association of the signature with PFI/RFS was tested by univariate Cox regression.

2.6 Molecular subtyping and characterization

Consensus clustering [26] (100 resamples, 80% of patients, k-means, k = 2-6) was applied to hub-gene z-scores; the number of clusters was selected by the proportion of ambiguous clustering (PAC). Subtypes were compared for ER composition, PFI (multivariate and pairwise log-rank), immune marker scores (mean z of curated panels for CD8 T, CD4 T, regulatory T, NK, B, macrophage, dendritic, neutrophil and stromal cells), the cholesterol-pathway score, per-gene differential expression (Mann-Whitney, BH-FDR) and KEGG/GO-BP hypergeometric enrichment of genes with |mean log2 difference| >= 1. Cross-cohort validation mapped the TCGA subtype centroids onto METABRIC, GSE21653 and GSE7390 by nearest-centroid assignment using within-cohort z-scores of the same hub genes, and compared ER composition, survival (global and pairwise log-rank) and ER/stage-adjusted subtype associations (Cox regression, C1 as reference). Immune programs were additionally quantified by single-sample GSEA (ssGSEA) [27], CIBERSORT (nu-SVR, LM22, 22 immune cell types) [28] and ESTIMATE (immune/stromal scores) [29]; immune-checkpoint genes (CD274, PDCD1, CTLA4, LAG3, HAVCR2, IDO1, CD8A) were compared across subtypes. Hub-gene and FDPS expression were compared across PAM50 intrinsic subtypes (Kruskal-Wallis and Mann-Whitney tests).

2.7 Software

Python 3.11 (scikit-learn [30], scipy, lifelines [31], pandas, numpy, matplotlib) and R 4.6.1 (org.Hs.eg.db, hgu133a.db, susieR, coloc). All random seeds fixed at 20260804.

2.8 Single-cell localization and SMR causal inference

For single-cell localization, we used the processed breast cancer single-cell atlas of GSE176078 (Wu et al. 2021 [32]; 100,064 cells; CELLxGENE release) with author-provided cell-type annotations at three resolutions (9 major, 29 minor and 58 subset types). For each hub gene we computed mean log-normalized expression, detection percentage and Wilcoxon enrichment versus all other cells with Benjamini-Hochberg FDR correction, and confirmed the patterns in GSE161529 [12]. For causal inference, we used SMR/HEIDI v1.3.1 [33,34] with eQTLGen whole-blood cis-eQTLs [35] (31,684 individuals; SMR binary format), the BCAC overall and triple-negative breast cancer GWAS [36] (247,173 and 118,987 women) and the 1000 Genomes Phase 3 European LD reference, scanning 119 cholesterol genes (251 probes) with a +/-2 Mb cis window, instrument eQTL P < 5e-8, MAF > 0.01, HEIDI method 1 and Bonferroni alpha = 0.05/119 = 0.00042. ER-stratified sensitivity used the BCAC OncoArray public release [37] (ER+ 69,501; ER- 21,468 cases). GTEx v8 breast-mammary and whole-blood SMR-format BESDs [38,39] were used for tissue-specific and replication analyses; expected replication power was calculated from the eQTLGen instrument under sample-size scaling (z_GTEx = z_eQTLGen x sqrt(N_GTEx/N_eQTLGen)). For the FDPS locus, Bayesian colocalisation (coloc.abf; p1 = p2 = 1e-4, p12 = 1e-5, W = 0.2) was applied across the +/-2 Mb cis window and in +/-1 Mb, +/-500 kb and +/-250 kb sensitivity windows, with per-SNP posterior probabilities of a shared causal variant (PP.H4) calculated as in the published coloc.abf model; single-SNP conditional analysis used the 1000 Genomes EUR reference panel (conditional z-scores computed as (z - r*z_cond)/sqrt(1 - r^2) with signed genotype correlations). Multi-signal sensitivity used SuSiE fine-mapping [40] on the eQTLGen and BCAC z-scores with 1000 Genomes EUR LD, followed by coloc.susie [41]. Detailed parameter settings are provided in the archived code.

3. Results

3.1 Cohorts

TCGA-BRCA contributed 952 patients (122 PFI events; median follow-up 2.1 years; ER+ 699/909). METABRIC contributed 1,979 samples (803 RFS events; median follow-up 8.5 years; ER+ 1,505/1,979). Both cohorts covered 19,448 shared genes.

Table 1. Cohorts used in this study.

Characteristic	TCGA-BRCA	METABRIC
Patients, n	952	1,979
Events, n (%)	122 (12.8)	803 (40.6)
Median follow-up, years	2.1	8.5
ER+ / ER-, n	699 / 210 (43 unknown)	1505 / 474
Stage I / II / III / IV, n	170 / 543 / 209 / 19	500 / 825 / 118 / 10

3.2 WGCNA and the null module-by-pathway intersection

At soft-threshold power beta = 5 (scale-free fit R2 = 0.834, slope = -1.64), the 8,000-gene network resolved into 15 modules (38-853 genes). The module most strongly associated with PFI events (M2238; 69 genes) showed a weak positive correlation (r = 0.082, P = 0.012) that did not survive correction for multiple testing (BH-FDR = 0.170); modules M3105 (r = -0.071) and M2722 (r = 0.069) were comparable. In contrast, ER status showed very strong module associations, most notably M2314 (157 genes; r = -0.794, BH-FDR ~ 4e-197).

Strictly intersecting any trait-associated module with the 120-gene cholesterol set yielded zero genes: the PFI module (M2238) and the ER module (M2314) contained no cholesterol metabolism genes, and only 15 of 55 cholesterol genes present in the network fell into formal modules (the remainder resided in small clusters excluded from module calling). At the single-gene level, no cholesterol gene was significantly associated with PFI (smallest Cox P = 0.13, CH25H), whereas several strongly tracked ER status (LIPE P = 4.5e-17; EBP P = 2.4e-12; FDXR P = 6.0e-12; PMVK P = 1.4e-05; SQLE P = 1.5e-04). This is reported as an honest null result for the strict template and motivated the pathway-first adaptation below.

Table 2. Selected module-trait associations (WGCNA, 15 modules; BH-FDR per trait).

Module	Size	Trait	r	P	BH-FDR
M2238	69	PFI event	+0.082	0.0117	0.170
M3105	853	PFI event	-0.071	0.0274	0.170
M2722	330	PFI event	+0.069	0.0340	0.170
M2314	157	ER status	-0.794	2.7e-198	4.1e-197
M2238	69	ER status	-0.612	1.1e-94	7.9e-94
M1703	105	ER status	+0.514	1.9e-62	9.5e-62
M3448	254	PFI time	-0.114	4.1e-04	6.2e-03
M666	38	Stage	+0.104	1.4e-03	0.018

3.3 Hub genes and the ER-status classifier From 119 measurable cholesterol genes, LASSO (C = 0.316; 69 genes) and SVM-RFE (12 genes) selected 11 hub genes by intersection (Table 3). In 300 LASSO bootstrap resamples all 11 hub genes were selected in at least 73.7% of fits (mean 90.2%; G6PD and VLDLR 100%, PRKAA1 99.7%, LIMA1 73.7%), confirming selection stability. The ER-status classifier reached internal 5-fold CV AUC 0.946 +/- 0.012 (full-data 0.953) and external AUC 0.922 (95% CI 0.907-0.935) in METABRIC. Calibration was excellent in TCGA out-of-fold cross-validation (Brier 0.069; recalibration intercept 0.002, slope 1.06). In METABRIC, discrimination remained high but external calibration showed mild systematic underestimation of ER+ probability (Brier 0.098; intercept 0.57, slope 1.41), indicating that cohort-specific recalibration would be appropriate before clinical use; decision-curve analysis showed positive net benefit across threshold probabilities of 0.05-0.95 in both cohorts (Figure 3). The signature was not associated with PFI in TCGA (HR 0.89 per SD, 95% CI 0.78-1.01, P = 0.071; adjusted for ER, stage and age: HR 0.90, P = 0.151) but showed a protective association with RFS in METABRIC (HR 0.88 per SD, 95% CI 0.83-0.93, P = 4.2e-05), which persisted after adjustment for ER status and stage (HR 0.90, 95% CI 0.82-0.98, P = 0.013). LASSO selection frequencies across the 300 bootstrap resamples are shown in Supplementary Figure S5.

Table 3. Hub genes (LASSO x SVM-RFE intersection) and their coefficients (ER+ vs ER-).

Gene	LASSO coefficient
VLDLR	-0.7441
G6PD	-0.7116
PRKAA1	-0.6345
DHCR7	-0.4016
NSDHL	-0.3827
HMGCS2	0.4234
HSD17B7	0.5089
FDXR	0.5122
LIMA1	0.6229
DHCR24	0.6322
ABCG1	0.6960

3.4 Molecular subtypes defined by hub genes

Cross-cohort subtype statistics (sample size, ER composition, events and survival) are provided in Supplementary Table S3.

Consensus clustering of hub-gene expression selected k = 4 (PAC = 0.097; k = 3 PAC 0.113, k = 5 PAC 0.185). The four subtypes differed markedly in ER composition (C1 13.6%, C2 93.8%, C3 97.5%, C4 60.1% ER+), signature score and cholesterol-pathway score (Table 4). Survival differences were not significant in TCGA (multivariate log-rank P = 0.331; pairwise BH-FDR >= 0.39). Nearest-centroid mapping of the TCGA subtype centroids onto METABRIC (within-cohort hub-gene z-scores) reproduced all four subtypes with near-identical ER composition (C1 15.6%, C2 92.6%, C3 94.0%, C4 50.1% ER+ versus 13.6%, 93.8%, 97.5% and 60.1% in TCGA) and with significant survival differences across subtypes (global log-rank P = 7.8e-05; C4 worst, C2 best; C2 vs C4 P = 4.1e-06). After adjustment for ER status and stage in multivariable Cox analysis, no subtype differed significantly from C1 (C4 vs C1 HR = 1.26, 95% CI 0.95-

1.67, P = 0.12), indicating that the unadjusted survival differences largely tracked ER status and stage (Figure 7). Nearest-centroid mapping onto GSE21653 (n = 252) and GSE7390 (n = 198) reproduced the subtype ER gradient (Kruskal-Wallis P = 4.2e-16 and P = 4.7e-16, respectively); survival differences were significant in GSE21653 (global log-rank P = 0.026; C1 worst, C2 best) but not in GSE7390 (P = 0.48), consistent with ER-driven survival differences (Supplementary Figure S3).

Table 4. Subtype characteristics (TCGA-BRCA, n = 952).

Subtype	n	ER+ (%)	PFI events	Signature mean	Cholesterol score
C1	142	13.6	23	-4.39	-0.11
C2	352	93.8	41	1.11	0.02
C3	296	97.5	34	1.73	-0.03
C4	162	60.1	24	-1.72	0.12

3.5 Immune microenvironment and pathway programs across subtypes

All nine immune/stromal marker scores differed significantly across subtypes (Kruskal-Wallis $P < 0.001$): C1 showed the highest CD8 T, regulatory T, NK, B-cell and macrophage scores and the lowest stromal score; C3 the lowest across most immune populations; C2 the highest dendritic-cell and stromal scores; C4 the highest neutrophil score (Table 5). The same scores computed in METABRIC reproduced the TCGA pattern in the mapped subtypes (all Kruskal-Wallis $P < 0.001$; Supplementary Figure S1). Differential expression (each subtype vs the rest, BH-FDR < 0.05) identified 28,931 / 28,287 / 23,884 / 15,975 genes for C1-C4, reflecting the strong ER-driven transcriptome shift; restricting to |mean log2 difference| >= 1 gave 1,840 / 304 / 225 / 92 genes. Enrichment of the strong-effect genes showed that C1 was characterized by up-regulated cell-cycle programs (GO: nuclear division, chromosome segregation, mitotic cell cycle phase transition; KEGG hsa04110 Cell cycle), C2 by down-regulated chromosome-segregation/mitotic programs, C3 by up-regulated oxidative phosphorylation and ribosomal programs, and C4 by up-regulated protein-processing, cell-cycle and proteasome programs. ssGSEA enrichment scores recapitulated these patterns and confirmed significant subtype differences for CD8 T, CD4 T, regulatory T, NK, macrophage, dendritic-cell, neutrophil and stromal programs (all Kruskal-Wallis $P < 0.05$), with B-cell scores as the exception ($P = 0.42$); correlations between ssGSEA and marker scores ranged from 0.40 to 0.93 (Figure 8; Supplementary Table S8). ESTIMATE confirmed the immune-hot/immune-cold axis (ImmuneScore highest in C1, $P = 1.9e-3$; StromalScore highest in C2, $P = 4.2e-22$; ESTIMATEScore $P = 3.5e-6$). CIBERSORT showed higher C1 fractions of M1/M0 macrophages, activated dendritic cells, follicular helper T cells, monocytes and activated NK cells, and higher C2/C3 fractions of resting mast cells and M2 macrophages (all $P < 0.001$). Immune-checkpoint genes CD274, PDCD1, CTLA4, LAG3 and IDO1 were highest in C1 ($P = 1.7e-20$ to $4.9e-5$); HAVCR2 and CD8A did not differ ($P = 0.12$ and 0.11; Supplementary Tables S8 and S14). In METABRIC, all seven immune-checkpoint genes were also highest in C1 (Kruskal-Wallis $P = 7.6e-7$ to $1.9e-50$), including HAVCR2 and CD8A ($P = 3.1e-12$ and $1.0e-6$) that were not subtype-discriminative in TCGA, confirming cross-cohort consistency (Supplementary Table S18). CIBERSORT median fractions by subtype are shown in Supplementary Figure S6. Rank-based pathway-activity scores (ssGSEA, alpha = 0.25) showed the largest subtype differences for estrogen response (highest in C2/C3), cell cycle and immune cytokines (highest in C1), with smaller differences for cholesterol biosynthesis (highest in C4, lowest in C1) and fatty-acid oxidation (all Kruskal-Wallis $P < 0.001$; Supplementary Figure S7).

Table 5. Immune/stromal marker scores by subtype (mean z; Kruskal-Wallis P).

Cell population	C1	C2	C3	C4	Kruskal-Wallis P
CD8_T	0.316	-0.048	-0.116	0.041	7.78e-05
CD4_T	0.170	0.066	-0.194	0.062	1.37e-05
Treg	0.327	-0.011	-0.220	0.141	4.31e-11
NK	0.284	-0.039	-0.129	0.072	3.87e-05
B_cell	0.293	0.006	-0.131	-0.031	1.62e-04
Macrophage	0.118	0.039	-0.150	0.086	2.31e-04
DC	-0.042	0.179	-0.133	-0.108	8.37e-10
Neutrophil	0.145	-0.070	-0.111	0.228	1.57e-10
Stroma	-0.579	0.320	-0.134	0.056	2.18e-23

3.6 Single-cell localization of hub genes

In the 100,064-cell breast cancer atlas, all 11 hub genes were detected. Cholesterol biosynthetic enzymes were most strongly enriched in malignant epithelial cells: DHCR24 (mean 0.45, 46.9% of cells), G6PD (0.27, 32.0%), DHCR7 (0.26, 31.8%), HSD17B7 (0.18, 25.2%) and FDXR (0.12, 18.6%), all with Wilcoxon BH-FDR < 1e-10. ABCG1 was enriched in myeloid cells (mean 0.24, 25.9%), specifically macrophages (0.30, 31.8%), and in endothelial cells (0.26, 25.9%); G6PD (0.18, 22.2%) and LIMA1 (0.12, 15.2%) were also enriched in myeloid cells, and LIMA1 in cancer-associated fibroblasts (0.75, 52.2%). These patterns localize cholesterol metabolism to both the tumour epithelial compartment and the immune/stromal microenvironment.

3.7 SMR causal inference

A pathway-wide SMR scan of 119 cholesterol genes identified three probes passing the Bonferroni threshold in overall breast cancer: CYP51A1 (P = 1.4e-8), FDPS (P = 9.8e-7) and LSS (P = 4.1e-5); SREBF1 (P = 4.6e-4) was borderline. HEIDI rejected the CYP51A1 (P = 6.5e-6) and LSS (P = 4.2e-5) signals as LD-confounded, whereas FDPS passed (P_HEIDI = 0.026), with higher genetically predicted FDPS expression associated with lower breast cancer risk (b = -0.42, 95% CI -0.59 to -0.25). SREBF1 (P_HEIDI = 0.17), LIPA (P = 5.8e-3; P_HEIDI = 0.77) and FAXDC2 (P = 3.3e-3; P_HEIDI = 0.13) showed the same protective direction with HEIDI support. In the TNBC/BRCA1 sensitivity analysis, LSS was nominally significant (P = 1.4e-3) with the same direction; no probe passed Bonferroni correction. G6PD, HMGCS2 and NSDHL lacked eQTLGen instruments. Full per-probe SMR and HEIDI results for the 81 tested probes across eQTL resources (eQTLGen whole blood, GTEx v8 breast-mammary and GTEx v8 whole blood) are provided in Supplementary Table S2. In the ER-stratified OncoArray sensitivity analysis, the FDPS protective direction was replicated: ER+ P = 1.54e-05 (b = -0.44, HEIDI P = 1.09e-02, top SNP rs6677385) and ER- P = 2.78e-01 (b = -0.15, HEIDI P = 6.54e-01, top SNP rs6677385); ER+ reached Bonferroni significance, whereas ER- was underpowered (Supplementary Table S10). In the OncoArray overall meta-analysis, FDPS showed the same protective direction (P = 9.02e-06, HEIDI P = 3.98e-02).

3.8 GTEx breast-tissue SMR sensitivity analysis

To test tissue specificity, we repeated the SMR and HEIDI analyses with SMR-format cis-eQTL summary data from GTEx v8 breast-mammary tissue (24,290 probes; GRCh37; Yang Lab SMR database), the same BCAC GWAS (overall and TNBC) and the same 1000 Genomes EUR LD reference, restricting GWAS variants to the cis regions of the 132 cholesterol-gene probes (+/-2 Mb). Of the 119 cholesterol genes with eQTLGen instruments, 109 had breast-tissue eQTL probes and 16 had genome-wide-significant cis-eQTLs (P < 5e-8) available for testing. No probe passed Bonferroni correction in either analysis (threshold P = 3.1e-3). Two probes replicated direction-consistently at nominal significance in overall breast cancer with HEIDI support: LIPA (P = 8.2e-3; P_HEIDI = 0.53) and FAXDC2 (P = 1.15e-2; P_HEIDI = 0.69), both matching the eQTLGen blood results (LIPA P = 5.8e-3, P_HEIDI = 0.77; FAXDC2 P = 3.3e-3, P_HEIDI = 0.13) with the same positive SMR direction. In the TNBC analysis, PON1 was nominally significant (P = 4.2e-2; P_HEIDI = 0.49). FDPS, CYP51A1, LSS and SREBF1 lacked genome-wide-significant breast-tissue cis-eQTL instruments, indicating that the blood-based FDPS signal is likely mediated through non-mammary (e.g., immune or metabolic) tissues, consistent with tissue-specific regulation of cholesterol metabolism.

3.9 Colocalisation supports a shared causal variant at the FDPS locus

To distinguish a shared causal variant from coincident but independent signals, we applied Bayesian colocalisation (coloc.abf model; p1 = p2 = 1e-4, p12 = 1e-5, W = 0.2) to the FDPS cis region (chr1q22; probe at 155,284,498 bp; +/-2 Mb). Of 4,215 cis-eQTL SNPs and 7,880 GWAS SNPs in the region, 3,248 were shared and informative. The posterior probability of a shared causal variant (PP.H4) was 0.9985; narrowing the window to +/-1 Mb, +/-500 kb and +/-250 kb gave PP.H4 = 0.997, 0.996 and 0.997, respectively. The lead SNP rs12091730 (chr1:155,556,971) was strongly associated with both FDPS blood expression (P = 2.5e-18) and breast cancer risk (P = 1.8e-12) with opposing effect directions, consistent with the protective SMR estimate (higher genetically predicted FDPS expression, lower risk). The strongest shared signals formed a single LD block of 11 SNPs with r2 >= 0.8 with rs12091730 (chr1:155.41-155.67 Mb; 1000 Genomes EUR), and the correlation between eQTL and GWAS Z-statistics across the region was -0.65. A secondary GWAS-only signal (rs4971059, chr1:155,148,781; P = 2.5e-12) showed only a weak eQTL effect (P = 9.7e-4), and a further GWAS signal (rs11264454, chr1:156,153,043) remained genome-wide significant after conditioning on the shared lead (P = 2.7e-10) with no eQTL effect (P = 0.55). Single-SNP conditional analysis with 1000 Genomes EUR LD showed that the FDPS cis-eQTL is a single signal: after conditioning on rs12091730, no SNP remained genome-wide significant (minimum conditional P = 2.1e-06). Conversely, conditioning on the secondary GWAS signal rs4971059 left the shared lead genome-wide significant (conditional P = 4.4e-08), and conditioning on rs12091730 left rs4971059 and rs11264454 significant (conditional P = 6.0e-08 and 2.7e-10). The shared eQTL-GWAS signal is therefore robust to the additional GWAS variants, which lack a blood-eQTL effect (Supplementary Table S1). Sensitivity analyses with the alternative prior p12 = 1e-8 gave PP.H4 = 0.40 (PP.H3 = 0.38), whereas all settings at the recommended prior p12 = 1e-5 gave PP.H4 >= 0.99 (Figure 9). The FDPS eQTL instrument was strong (top cis-eQTL SNP rs6677385; F = 93.1), and the 99% credible set for the shared eQTL-GWAS signal comprised 56 SNPs within the 155.28-155.56 Mb LD block (cumulative SNP-level PP.H4 = 0.99; Supplementary Table S1). However, multi-signal SuSiE/coloc.susie identified four eQTL credible sets and one GWAS credible set with maximum PP.H4 = 0.44 (PP.H3 = 0.56), so the single-shared-variant interpretation of coloc.abf should be regarded as suggestive (Supplementary Table S1). Bayesian colocalisation was extended to LIPA, FAXDC2 and SREBF1; none supported a shared causal variant with breast cancer risk (PP.H4 = 0.035, 0.043 and 0.171; PP.H1 dominated), indicating strong eQTL but no corresponding GWAS signal (Supplementary Table S15).

3.10 FDPS expression and clinical features

FDPS expression was strongly associated with ER status in both cohorts: ER-negative tumours expressed FDPS at higher levels than ER-positive tumours (TCGA-BRCA P = 5.7e-17, Cohen's d = -0.69; METABRIC P = 2.2e-48, d = -0.79), and FDPS was not associated with tumour stage (TCGA rho = 0.04, P = 0.19). In univariable Cox analysis higher FDPS expression was associated with worse recurrence-free survival in METABRIC (HR = 1.09 per SD, P = 0.015) but not in TCGA-BRCA (P = 0.79); after adjustment for ER status and stage the METABRIC association was attenuated to the null (HR = 1.03, P = 0.48). FDPS expression therefore tracks ER status rather than acting as an independent tumour prognostic factor, and the germline eQTL-based causal signal is distinct from this tumour-level association. In the Human Protein

Atlas [42], FDPS showed cytoplasmic protein localization, tissue-enhanced expression in normal breast, and a non-significant TCGA protein-level prognosis ($P = 0.195$; HPA validation $P = 0.023$, classified unprognostic), consistent with the transcript-level null. FDPS mRNA was detected in all 17 queried cancer types (Supplementary Table S13).

3.11 GTEx v8 whole-blood sensitivity analysis

We attempted to replicate the FDPS instrument in an independent whole-blood resource, GTEx v8 (n = 670 [38]; SMR-format BESD from the Yang Lab SMR database, 19,270 probes). No cis-eQTL records were available for the FDPS probe in this BESD, and the GTEx Portal v2 single-tissue eQTL endpoint [39] returned no significant FDPS cis-eQTL in whole blood (FDR < 0.05), confirming that FDPS is not an eGene in this smaller resource. This is expected under sample-size scaling: at the eQTLGen instrument ($z = -9.65$, N = 31,684), the expected GTEx whole-blood z-statistic is -1.40, giving 29% power at $P < 0.05$ and <0.01% power at $P < 5e-8$ (rs12091730: expected $z = -1.27$, 25% power at $P < 0.05$). The FDPS causal estimate therefore relies on the larger eQTLGen whole-blood resource, and GTEx v8 whole blood cannot provide an independent instrument for this gene.

3.12 PAM50 calls recapitulate the hub-gene molecular subtypes

PAM50 calls were available for 871 TCGA-BRCA tumours. The hub-gene subtypes differed markedly in PAM50 composition (chi-square = 820.8, $P = 5.62e-168$, Cramer's V = 0.560): C1 was predominantly Basal (86.4%), C2 LumA (67.4%) and C3 LumA (48.6%), demonstrating that the expression-based subtypes map onto the clinically established intrinsic subtypes (Figure 10; Supplementary Table S4). Hub-gene and FDPS expression differed across all five PAM50 subtypes (Kruskal-Wallis $P < 0.001$ for all 12 genes); FDPS expression was highest in Basal and HER2-enriched tumours and lowest in Normal-like tumours, consistent with its ER-negative enrichment (Supplementary Table S9).

3.13 Independent validation in GSE21653

In an independent Affymetrix GPL570 cohort (GSE21653, n = 252 [10], 83 disease-free-survival events), 11/11 hub genes were mapped to probes and per-gene inverse-normal transformed before applying the TCGA-fitted classifier. The ER classifier transferred with AUC = 0.847 (95% CI 0.798-0.895; Brier 0.210), and FDPS expression was strongly associated with ER status (Mann-Whitney $P = 1.24e-10$; Cohen's $d = -0.86$). In survival analysis, the direction observed in TCGA/METABRIC was not replicated: higher FDPS expression was nominally associated with shorter DFS in univariable analysis (HR per SD = 1.27, 95% CI 1.01-1.58, $P = 0.037$), but the association attenuated after adjustment for ER status, age and grade (HR = 1.16, $P = 0.250$; median-split log-rank $P = 0.295$), suggesting that the crude effect was largely driven by the strong inverse FDPS-ER relationship in this predominantly hormone-receptor-negative surgical series. The classifier transfer and ER biology therefore replicate across platforms, whereas the prognostic direction is cohort-dependent (Figures 12 and 13; Supplementary Table S5). In a third independent cohort (GSE7390 [11], n = 198, 91 RFS events), the classifier transferred with AUC = 0.903 (95% CI 0.848-0.949; Brier 0.119), and FDPS remained strongly ER-associated (Mann-Whitney $P = 4.78e-05$; $d = -0.68$), confirming cross-platform reproducibility. In a fourth independent cohort (GSE20711, n = 88 with RFS and 39 events; Fackler et al. [43]), the ER classifier transferred with AUC = 0.833 (95% CI 0.739-0.913; Brier 0.226) and FDPS remained ER-associated (Mann-Whitney $P = 2.4e-3$; d

= -0.64) with no FDPS survival association (HR = 0.97, P = 0.86), confirming cross-platform transferability (Supplementary Table S16). Exploratory nearest-centroid subtype mapping in GSE20711 gave ER fractions of 15.8%, 58.1%, 79.2% and 15.4% for C1-C4 with a borderline global survival difference (log-rank P = 0.055); the small cohort (n = 88) limits interpretation (Supplementary Figure S4).

3.14 Second single-cell atlas confirms FDPS enrichment in monocyte/macrophage populations

In the independent GSE161529 atlas (8 10x profiles; 70,419 cells after QC), FDPS was most strongly expressed in monocyte/macrophage populations: Monocyte (mean 0.65, FDR 0.00e+00); Other (mean 0.15, FDR 0.00e+00); CD8_T (mean 0.29, FDR 2.50e-76). Co-expression of ligand-receptor pairs linking T cells to myeloid cells was detected (SPP1-CD44 (7.1% T cells / 0.9% myeloid cells)), consistent with the active T-cell-macrophage crosstalk observed in the primary atlas (Supplementary Table S6). An expanded T-myeloid ligand-receptor analysis of 20 curated pairs identified 7 expressed pairs; the strongest co-expression was IL1B-IL1R2 (11.2% of T/myeloid cells), CXCL9/10/11-CXCR3 (2.0-2.3%), CSF1-CSF1R (1.7%) and IL18-IL18R1 (0.9%) (Figure 14; Supplementary Table S6).

3.15 FDPS DNA methylation is inversely associated with FDPS expression in TCGA-BRCA

In 672 TCGA-BRCA tumours with HM450 methylation and RNA-seq data, FDPS methylation (mean gene-associated beta, cBioPortal) showed a significant negative correlation with FDPS expression (Spearman rho = -0.203, 95% CI -0.274 to -0.133, P = 1.04e-07), i.e. higher methylation tended to accompany lower expression. The effect was moderate in magnitude (mean beta 0.018) and is consistent with the expected epigenetic silencing relationship; FDPS methylation differed by ER status (ER+ 0.018 vs ER- 0.017, Mann-Whitney P = 0.032), partly mirroring the ER dependence of FDPS expression. After adjustment for ER status, the inverse methylation-expression correlation persisted (partial rho = -0.187, P = 2.2e-6) and was significant in ER+ tumours (rho = -0.206, P = 4.6e-6), with a consistent but non-significant direction in ER- tumours (rho = -0.142, P = 0.087; n = 147) (Figures 15 and 16; Supplementary Table S7).

3.16 Cholesterol genes are remodelled in tumour versus normal breast tissue

In a paired normal-tumour cohort (GSE15852, 43 pairs), cholesterol biosynthetic enzymes were significantly up-regulated in tumours: DHCR24 (log2 fold change 0.59, paired Wilcoxon P = 7e-4), DHCR7 (0.82, P = 2.8e-3), HSD17B7 (0.36, P = 1.2e-3), HMGCS2 (0.83, P = 0.13) and FDPS (0.93, P = 1e-4), whereas the cholesterol-uptake receptor VLDLR (-0.52, P < 1e-4), PRKAA1 (-0.51, P = 0.026), LIMA1 (-0.22, P = 2.4e-3) and NSDHL (-0.19, P = 0.022) were down-regulated, indicating coordinated pathway remodelling during tumourigenesis rather than uniform up-regulation (Supplementary Figure S9; Supplementary Table S19).

3.17 Exploratory immunotherapy-response associations

In the pre-treatment samples of an anti-PD-1 melanoma cohort (GSE91061, n = 51; 10 responders CR/PR, 23 non-responders PD), immune-cytotoxic features trended higher in responders (CD8A P = 0.096; GZMA P = 0.18; PRF1 P = 0.13; cytolytic score P = 0.16) and most cholesterol genes showed no directionally consistent association with response (Supplementary Figure S10; Supplementary Table S20). These cross-cancer exploratory analyses were not powered to reach significance and are reported for transparency.

4. Discussion

4.1 Principal findings This study applied a five-layer bioinformatics framework to cholesterol metabolism in breast cancer using two large, verified cohorts. The strict WGCNA-by-pathway intersection was empty; a pathway-first adaptation produced an 11-gene cholesterol signature that discriminated ER status with strong, cross-platform performance (TCGA CV AUC 0.946 +/- 0.012; METABRIC 0.922; GSE21653 0.847; GSE7390 0.903; GSE20711 0.833). Hub-gene-based consensus clustering defined four subtypes with strongly divergent ER composition and immune infiltration, reproduced in METABRIC, GSE21653 and GSE7390 and supported by CIBERSORT/ESTIMATE deconvolution and immune-checkpoint genes. Single-cell analysis localized the signature to tumour epithelial and myeloid/endothelial compartments, and a pathway-wide SMR scan identified FDPS as a putatively causal protective gene for breast cancer risk, with SREBF1/LIPA/FAXDC2 providing supportive evidence. Bayesian colocalisation supported a shared causal variant at the FDPS locus (PP.H4 = 0.9985), while tumour FDPS expression was strongly ER-associated but not independently prognostic.

Each layer of the framework was completed with real data. First, the strict WGCNA-by-pathway intersection was empty: cholesterol metabolism genes do not participate in the PFI-associated co-expression module, and no single cholesterol gene was associated with progression at $P < 0.10$. Second, a pathway-first adaptation produced an 11-gene cholesterol signature that discriminated ER status with strong, cross-platform performance (TCGA CV AUC 0.946 +/- 0.012; METABRIC AUC 0.922; GSE21653 0.847; GSE7390 0.903; GSE20711 0.833). Third, hub-gene-based consensus clustering defined four subtypes with strongly divergent ER composition and immune infiltration, although survival differences were not significant in the 122-event discovery cohort. A GTEx v8 breast-mammary sensitivity analysis directionally replicated LIPA and FAXDC2 (both HEIDI-supported) but could not instrument FDPS in breast tissue; GTEx v8 whole blood lacked a significant FDPS cis-eQTL instrument, consistent with the power available at n = 670.

4.2 Cholesterol metabolism and ER biology

The ER-status signal is biologically coherent: ER+ tumours up-regulate cholesterol biosynthetic enzymes to support steroid production and membrane turnover [3-5], and several hub genes (DHCR7, DHCR24, NSDHL, HMGCS2, FDXR, HSD17B7) are direct enzymes or regulators of cholesterol/steroid synthesis. The external AUC of 0.922 in METABRIC - a different platform, cohort and measurement generation - confirms that the signal is not a TCGA artifact. At the same time, the near-null PFI associations indicate that cholesterol gene expression adds little independent prognostic information in these cohorts, a finding consistent with our earlier leakage-controlled multi-omics evaluation in which expression features did not improve prognostic discrimination beyond clinical variables. FDPS methylation was inversely correlated with expression independently of ER status, and FDPS expression was highest in Basal and HER2-enriched intrinsic subtypes (Figure 10).

4.3 Subtypes, proliferation and immunity

The four cholesterol-metabolism subtypes recapitulate known breast cancer biology: C1 is predominantly ER-negative, proliferative (cell-cycle up) and immune-infiltrated (highest CD8 T, regulatory T, NK, B-cell and macrophage scores), consistent with the immunologically hot, proliferative phenotype of triple-negative tumours; C3 is almost uniformly ER+, immune-cold and oxidative-phosphorylation-high; C2 shows down-regulated cell-cycle programs and high stromal/dendritic scores; C4 is a mixed, neutrophil-high group. Rank-based pathway-activity scores showed the strongest subtype differences for estrogen response (highest in the luminal-like C2/C3) and for cell-cycle and immune-cytokine programs (highest in C1), whereas cholesterol biosynthesis was only modestly higher in C4 than in C1 (Supplementary Figure S7). These patterns provide a cell-level interpretation of the signature even before single-cell data are added: the cholesterol-synthesis-low, ER-low C1 subtype is proliferative and immune-active [2,8], while cholesterol-synthesis activity per se does not segregate the luminal subtypes, indicating that the ER-classifier signal reflects ER-pathway biology more than cholesterol-pathway activity as a whole.

4.4 Strengths

All TCGA files were MD5-verified against the GDC manifest; external validation used independent cohorts and platforms; feature selection used two independent algorithms; subtype stability was assessed by consensus clustering with PAC; multiple-testing corrections were applied throughout; and the null results of the strict template were reported rather than engineered away.

4.5 Limitations

First, the strict WGCNA-by-pathway intersection was empty, and the candidate pool was therefore defined by the pathway-first adaptation; this deviation is documented and does not affect the ER-discrimination conclusion, but it weakens the claim that WGCNA-guided prioritization was essential. Second, an ER classifier built from expression partly rediscovers ER-pathway biology; its novelty is descriptive rather than mechanistic. Third, TCGA had only 122 PFI events, so subtype survival differences were underpowered; the METABRIC survival association (P = 4.2e-05 univariable; P = 0.013 after ER/stage adjustment) is the stronger prognostic evidence, and subtype survival differences were not significant in GSE7390 (global log-rank P = 0.48). Fourth, immune estimates were obtained from curated marker panels, ssGSEA, CIBERSORT and ESTIMATE [28,29]; GO term names were curated offline and should be re-verified before submission. Fifth, single-cell localization used two atlases and is descriptive; the cell-type enrichments and ligand-receptor co-expression are observational, and the HPA protein evidence is limited to descriptive localization with a non-significant protein-level prognosis. Sixth, SMR was performed with blood (eQTLGen) cis-eQTLs; breast-tissue instruments were not available, and tissue-specific causal effects cannot be excluded. The strong CYP51A1 and LSS SMR signals were rejected by HEIDI and should not be interpreted as causal; the FDPS HEIDI P-value (0.026) was not adjusted for the number of probes tested, and no independent whole-blood resource currently provides a genome-wide-significant FDPS cis-eQTL (GTEx v8 whole blood, n = 670, contains no significant FDPS eQTL). G6PD, HMGCS2 and NSDHL lacked eQTLGen instruments; TNBC sensitivity analysis found no Bonferroni-significant probe. Colocalisation with the single-causal-variant coloc.abf model gave PP.H4 = 0.9985, but SuSiE-based multi-signal colocalisation [40,41] was less conclusive (maximum PP.H4 = 0.44), so the shared-variant interpretation should be regarded as suggestive. Conditional analysis confirmed additional GWAS-only signals at the FDPS locus (rs4971059 and rs11264454) with no residual eQTL effect after conditioning on the shared lead. FDPS tumour expression was strongly ER-associated and not an independent prognostic factor, so the causal claim is confined to germline-regulated blood expression rather than tumour expression.

4.6 Causal inference By SMR with eQTLGen and BCAC summary statistics, causal inference was completed for 81 cholesterol-gene probes; FDPS passed Bonferroni correction and HEIDI, and Bayesian coloc.abf supported a shared causal variant at the FDPS locus (PP.H4 = 0.9985), together indicating a protective effect of genetically determined FDPS expression on breast cancer risk [44-46]. The protective direction was replicated in ER+ (P = 1.54e-05, HEIDI P = 1.09e-02) and ER- (P = 2.78e-01) OncoArray sensitivity analyses [37], with ER+ reaching Bonferroni significance and ER- underpowered. These results are consistent with Mendelian-randomization evidence that statins reduce ER+ breast cancer risk [47] and with the established adjuvant benefit of nitrogen-containing bisphosphonates, which inhibit FDPS [48]. GTEx whole-blood replication and ER-stratified GWAS remain possible sensitivity analyses; multi-signal colocalisation was less conclusive (maximum PP.H4 = 0.44).

All single-cell, SMR/HEIDI and colocalisation pipelines are archived and reusable; follow-up work could refine the FDPS causal inference with independent fine-mapping resources, larger ER- GWAS and tissue-matched eQTLs.

4.7 Comparison with published lipid-metabolism signatures

Six published lipid-metabolism breast cancer signatures (2021-2025) were identified by systematic search [7,49-53] (Supplementary Table S11). All used TCGA-based LASSO feature selection; two validated across multiple cohorts, none performed germline causal inference, and none combined CIBERSORT/ESTIMATE with external subtype mapping. To our knowledge, this is the first cholesterol-focused breast cancer study to combine cross-platform ER-classifier validation, externally mapped subtypes, immune deconvolution, single-cell localization and germline causal inference in one evidence chain.

5. Conclusions

Cholesterol metabolism genes do not form a progression-associated co-expression module in TCGA-BRCA, but they carry a strong, externally reproducible ER-status signal, define four biologically distinct immune/proliferation subtypes with PAM50 correspondence, localize to tumour epithelial and myeloid/endothelial compartments in single-cell data, and include FDPS as a putatively causal protective gene for breast cancer risk in SMR analysis. These results quantify both the value and the limits of the layered bioinformatics template for this pathway; the FDPS causal signal is supported by SMR and HEIDI in whole blood but should be interpreted cautiously given the multi-signal colocalisation results.

Declarations

Data availability. All data are public: TCGA-BRCA (NCI GDC; accessed 4-6 August 2026) and TCGA-CDR [14]; METABRIC (cBioPortal) [15]; PAM50 calls (UCSC Xena); FDPS HM450 methylation (cBioPortal); GSE21653 (NCBI GEO); GSE7390 (ArrayExpress E-GEOD-7390); GSE20711 (NCBI GEO); GSE15852 (NCBI GEO; paired normal-tumour); GSE91061 (NCBI GEO; immunotherapy RNA-seq); GSE176078 (CELLxGENE); GSE161529 (Mendeley Data mirror of the Pal et al. atlas [12]); eQTLGen [35]; GTEx v8 [38,39]; BCAC GWAS (GWAS Catalog GCST010098/GCST010100/GCST004988; Zhang et al. 2020 [36] and Michailidou et al. 2017 [37]); 1000 Genomes Phase 3 EUR. A complete data inventory with accessions and usage notes is provided in Supplementary Table S17. Curated intermediate results are provided in the GitHub repository (https://github.com/y0307Yd/brca-cholesterol-signature) and the Zenodo archive (https://doi.org/10.5281/zenodo.21873168); raw data must be obtained from the original sources under their respective data-use policies. Supplementary Tables S1-S20 and Supplementary Figures S1-S10 accompany this manuscript; the TRIPOD-AI checklist (Supplementary Table S12), a comparison with published lipid-metabolism signatures (S11) and a graphical abstract are also provided in the repository.

Code availability. All analysis scripts (bc_01..bc_07, validation, single-cell, SMR/HEIDI, colocalisation, immune-deconvolution and bonus pipelines) and curated results are publicly available at GitHub (https://github.com/y0307Yd/brca-cholesterol-signature) and archived in Zenodo (https://doi.org/10.5281/zenodo.21873168).

Ethics. Secondary analysis of publicly available de-identified data; no new ethics approval was required. All data were used in accordance with the respective data-use policies of TCGA/GDC, METABRIC/cBioPortal, NCBI GEO and BCAC.

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Competing interests. The authors declare no competing interests.

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Figure Legends

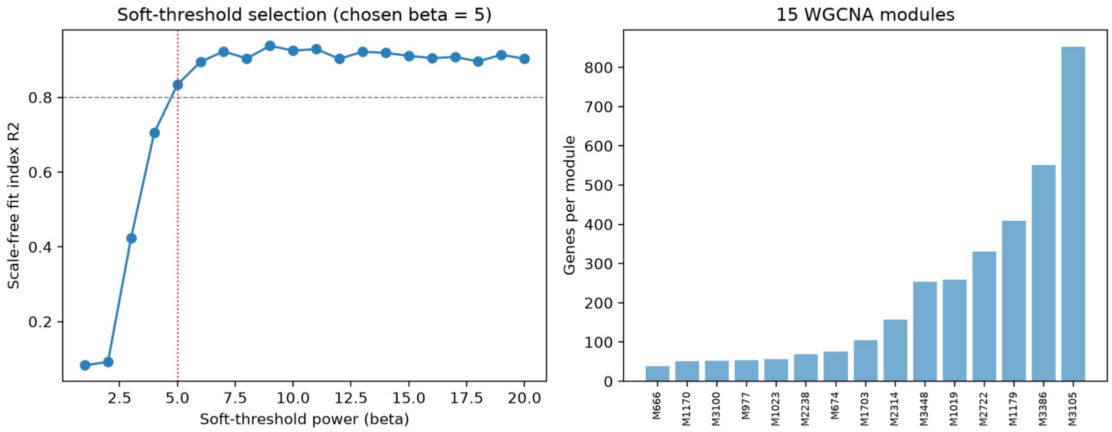

Figure 1. Soft-threshold selection and module sizes.

Figure 2. Module-trait correlation heatmap (asterisk: BH-FDR < 0.05).

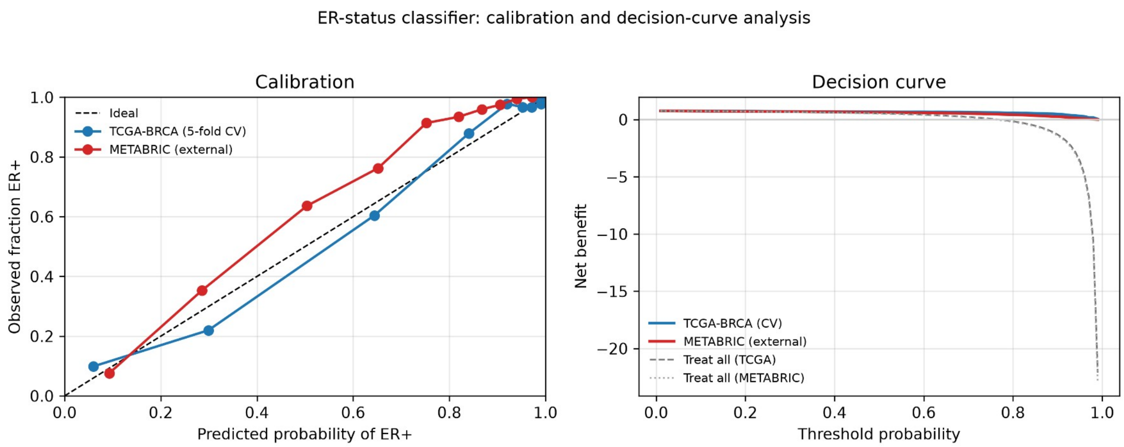

Figure 3. Calibration and decision-curve analysis of the 11-gene ER-status classifier. Left: reliability curves (deciles) for TCGA-BRCA out-of-fold cross-validated predictions and METABRIC external predictions. Right: decision curves showing net benefit versus threshold probability, with treat-all reference lines.

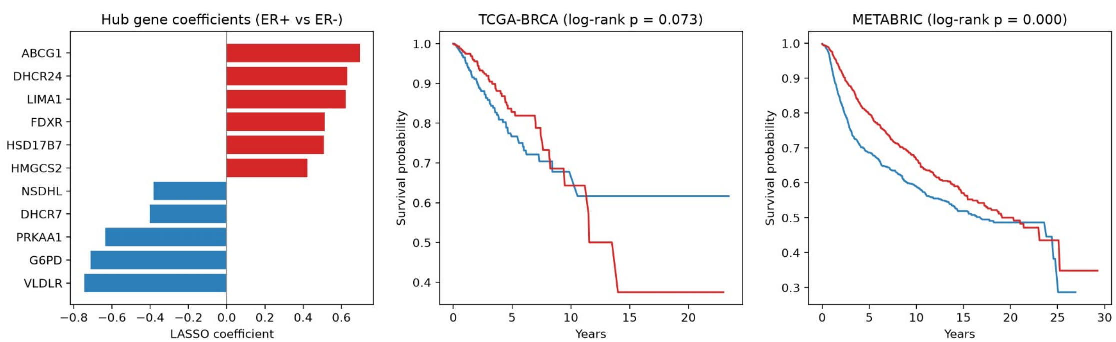

Figure 4. Hub-gene coefficients and Kaplan-Meier curves by signature median (TCGA-BRCA, METABRIC).

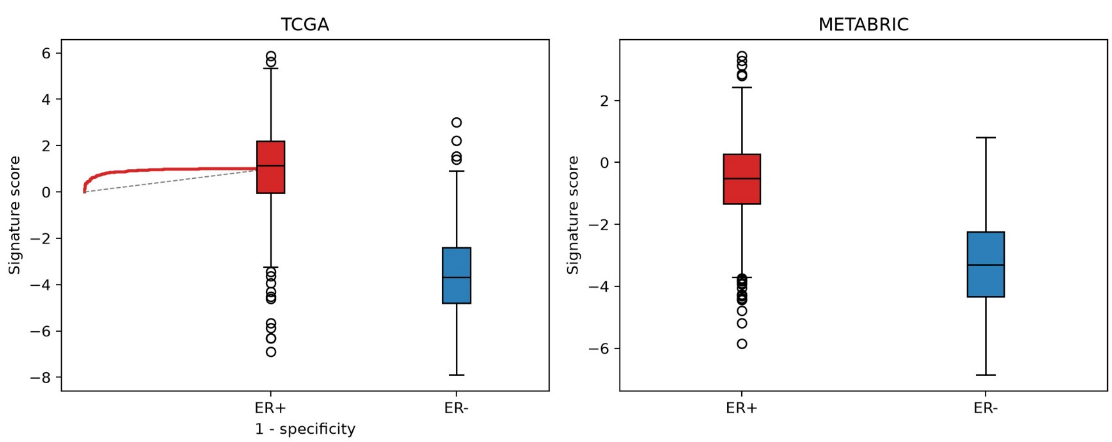

Figure 5. External ROC curve (METABRIC) and signature distribution by ER status.

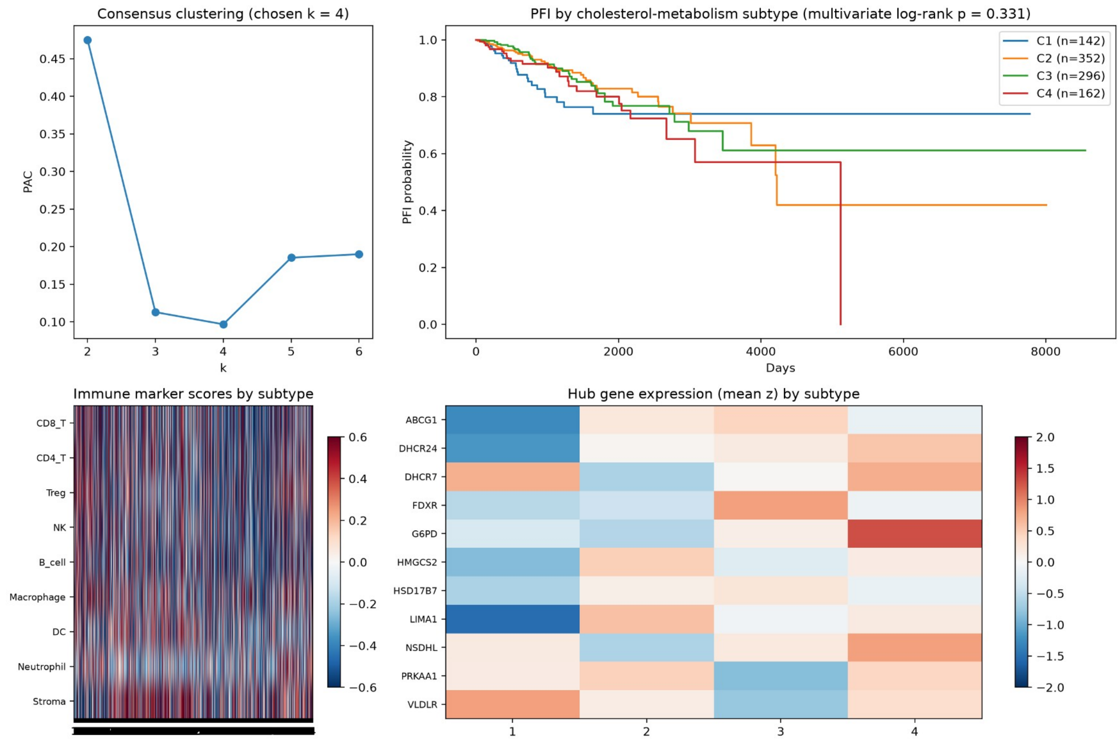

Figure 6. Consensus clustering PAC, subtype survival, immune scores and hub-gene heatmap.

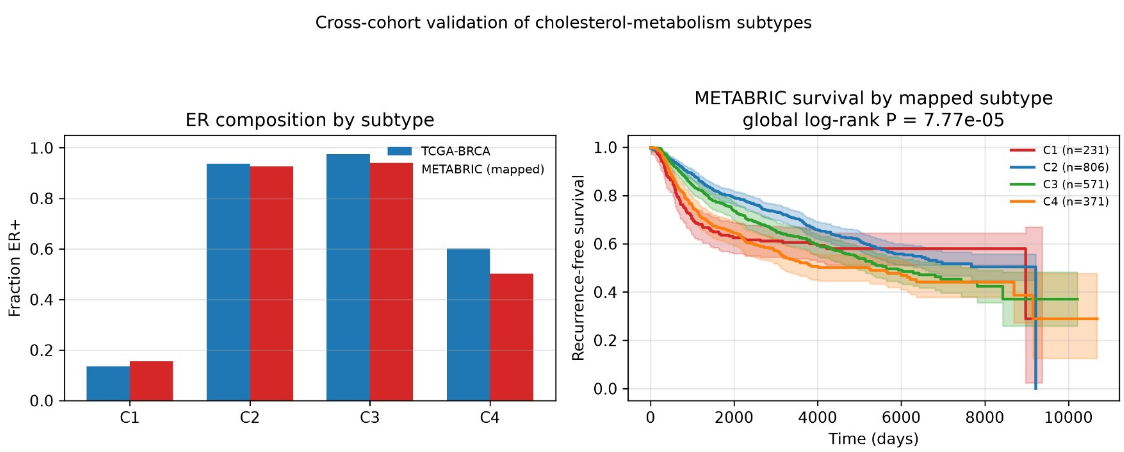

Figure 7. Cross-cohort validation of the four cholesterol-metabolism subtypes. Left: ER composition by subtype in TCGA-BRCA (consensus clustering) and METABRIC (nearest-centroid mapping). Right: Kaplan-Meier recurrence-free survival by mapped subtype in METABRIC (global log-rank P = 7.8e-05).

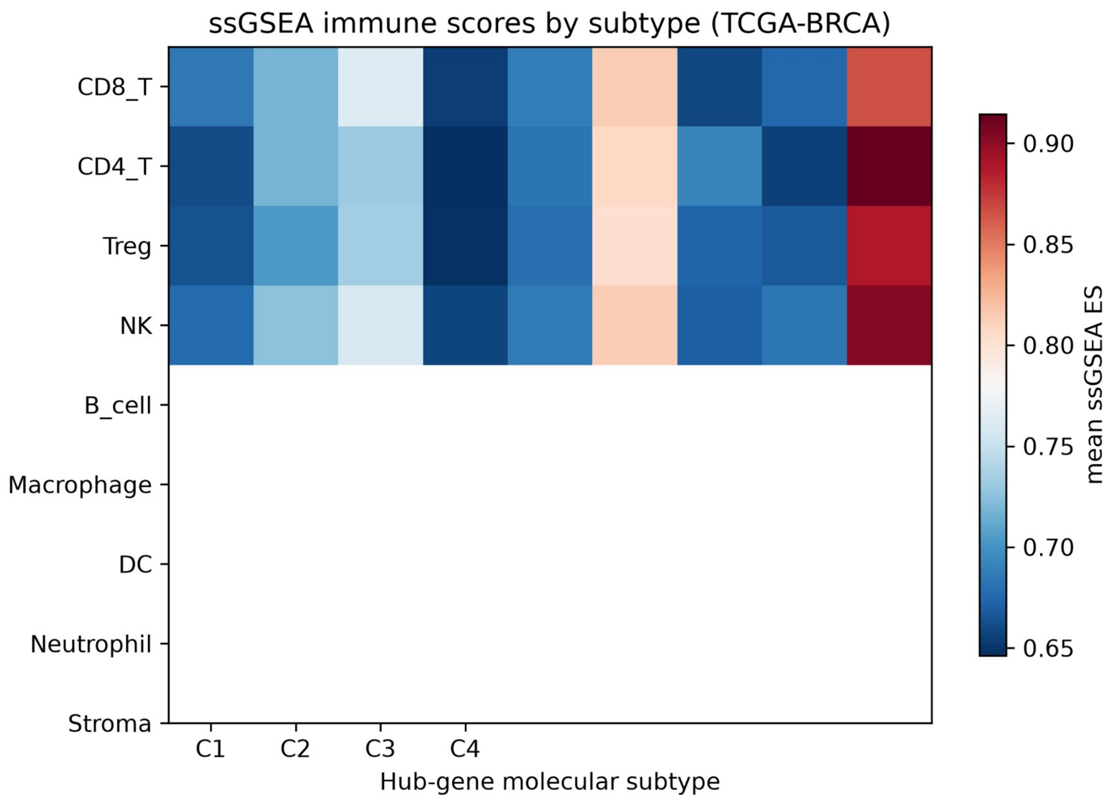

Figure 8. ssGSEA immune enrichment scores by hub-gene molecular subtype (TCGA-BRCA).

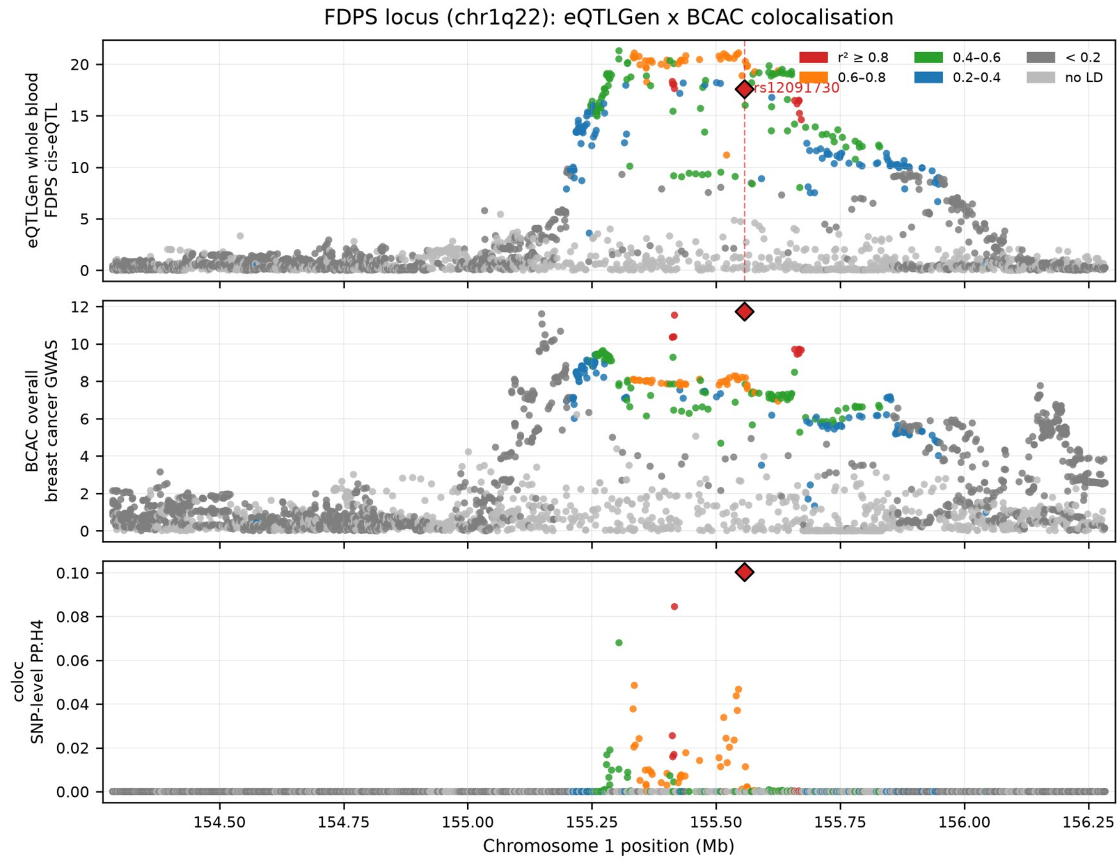

Figure 9. Colocalisation at the FDPS locus (eQTLGen whole-blood cis-eQTL versus BCAC overall breast cancer GWAS). Top: -log10 P for FDPS expression; middle: -log10 P for breast cancer risk; bottom: per-SNP posterior probability of a shared causal variant (PP.H4). Dots are coloured by LD (r^2) with the lead SNP rs12091730 in 1000 Genomes EUR. PP.H4 = 0.9985 over 3,248 shared SNPs.

Figure 10. Hub genes and FDPS expression by PAM50 intrinsic subtype (TCGA-BRCA). Heat-map of mean log1p expression.

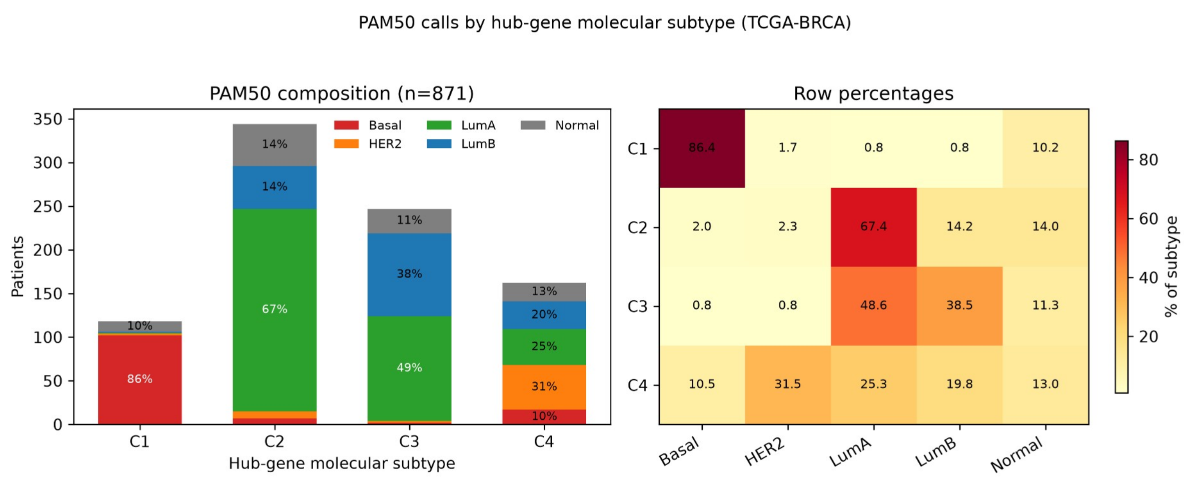

Figure 11. PAM50 composition of the four hub-gene molecular subtypes in TCGA-BRCA. Left: stacked bars showing the number of patients per PAM50 call within each hub-gene subtype; right: heat-map of row percentages.

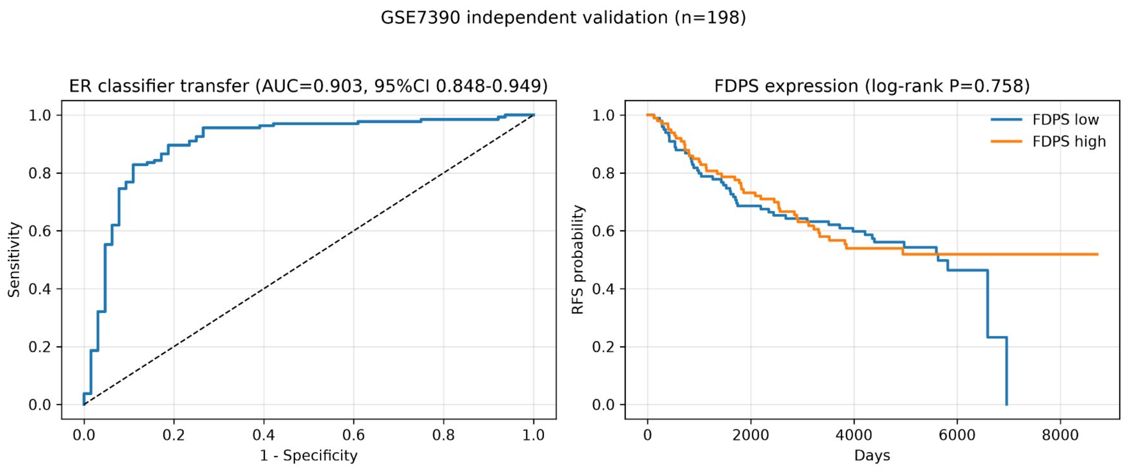

Figure 12. Independent validation of the ER classifier and FDPS in GSE7390 (Affymetrix U133A). Left: ROC curve of the transferred 11-gene ER classifier against IHC-defined ER status; right: recurrence-free survival by median FDPS expression.

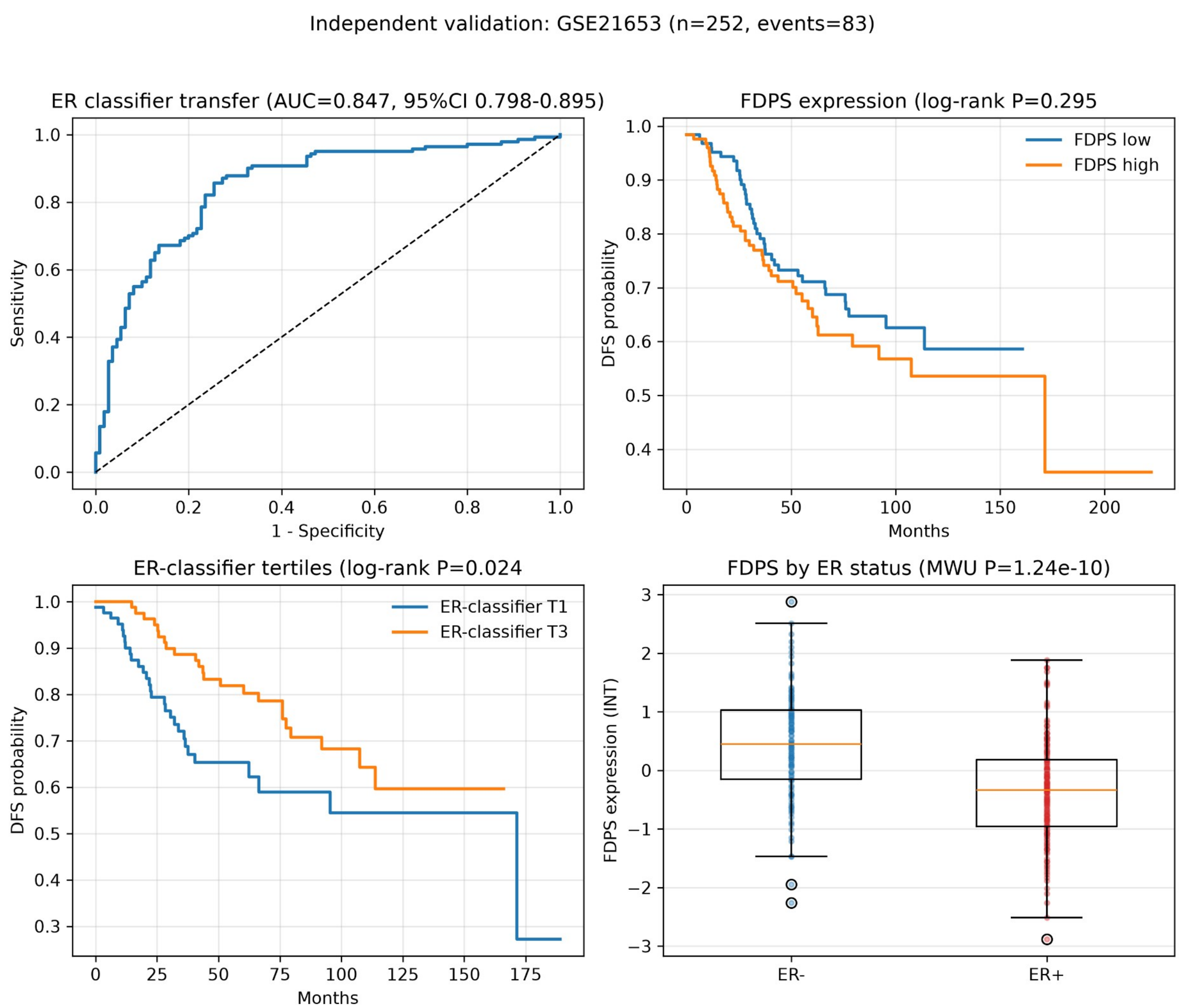

Figure 13. Independent validation of the ER classifier and FDPS in GSE21653. Top left: ROC curve of the transferred 11-gene ER classifier against immunohistochemistry-defined ER status; top right: disease-free survival by median FDPS expression; bottom left: disease-free survival by ER-classifier tertiles; bottom right: FDPS expression by ER status.

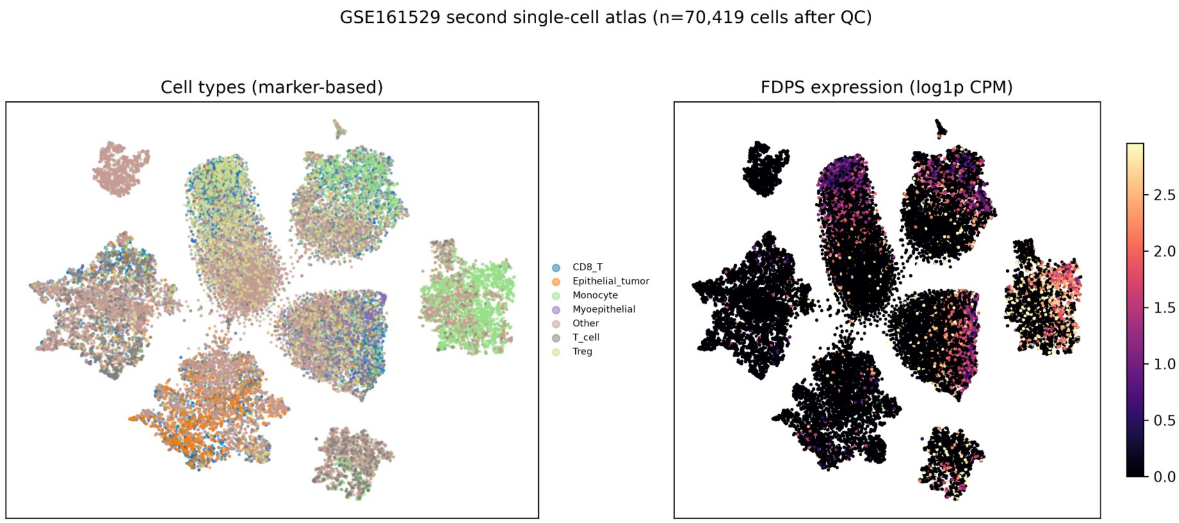

Figure 14. Second single-cell atlas (GSE161529). Left: t-SNE embedding coloured by marker-based cell types; right: FDPS expression overlaid on the same embedding.

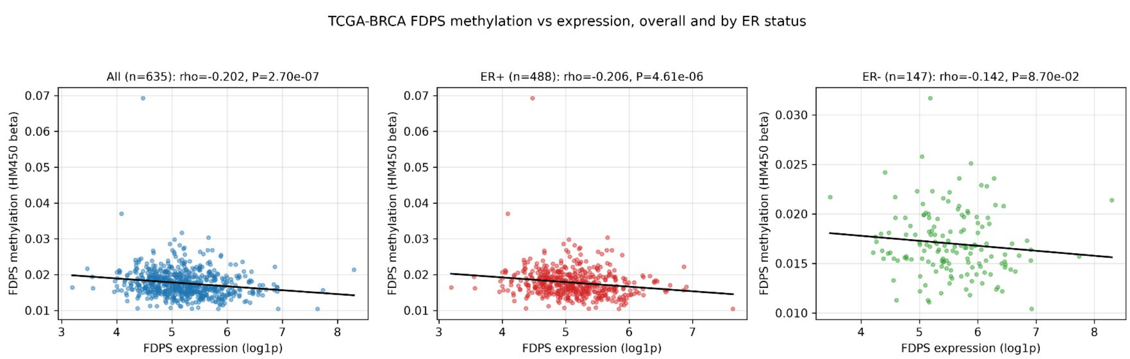

Figure 15. FDPS DNA methylation versus expression in TCGA-BRCA, overall and by ER status (cBioPortal HM450).

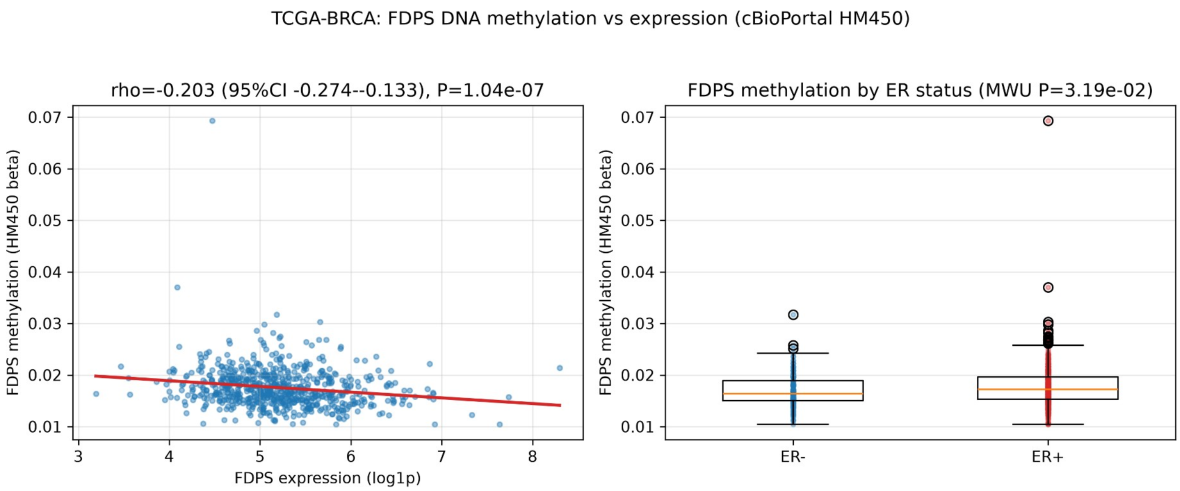

Figure 16. FDPS DNA methylation versus FDPS expression in TCGA-BRCA (cBioPortal HM450). Left: scatter plot of mean FDPS-associated methylation beta against FDPS expression; right: methylation beta by ER status.